Pediatric Physical Assessment

Physical examination



Third Stage

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Pediatric Nursing

Objectives

- - Define physical examination
- - Illustrate types of physical examination
- - Explain preparation of the environment (examination room)
- - Identify the sequence of pediatric physical examination

Definition

- - Physical examination: Essential part of pediatric health assessment
- - Requires strong physical assessment skills
- - Knowledge of child development is critical
- - Key techniques: Inspection, palpation, percussion, auscultation

Pediatric vs. Adult Examination

- - Pediatric exam differs by age & developmental level
- - Approach & sequence tailored to child's stage
- - Importance of comfort and cooperation (Chiocca, 2015)

Types of Physical Examination

- - Complete (Head-to-Toe): New patients, annual visits, school/sports physicals
- - Focused Examination: Based on chief complaint, performed during illness, injury, or emergency

Examination Techniques

- - Inspection
- - Palpation
- - Percussion
- - Auscultation
- - Abdomen sequence: Inspection → Auscultation → Palpation → Percussion

Use of Senses in Exam

- - Sight: color, respiratory effort, posture, gait
- - Smell: hygiene, infections
- - Touch: masses, skin changes
- - Hearing: body sounds

Preparation of the Child

- - Consider age & developmental level
- - Build rapport: play with infants, conversation with older children
- - Allow handling of equipment
- - Example: Listening to parent’s heart with stethoscope

Preparation of the Environment

- - Comfortable temperature, well lit, nonthreatening
- - Age-appropriate decoration
- - Keep instruments out of sight until needed
- - Childproof environment
- - Toys & reading material for distraction & assessment
- - Ensure privacy at all ages

Sequence of Examination by Age

- - Newborns (0–6 months): Supine, radiant warmer, diaper only, least distressing exam first
- - Older Infants (6–12 months): Parent’s lap if anxious, stranger anxiety begins at 7–8 months





Toddlers

- - Exam on parent’s lap or table
- - Use play & distraction
- - Stranger anxiety decreases ~18 months
- - Often negativistic → Offer simple choices
- - Praise cooperation



Preschoolers

- - Exam table with parent nearby
- - Use dolls/toys for demonstration
- - Provide gown, remove clothes except underwear
- - Concrete thinkers, magical thinking
- - Protect modesty, careful with word choices



School-aged Children





- - Exam table, provide gown
- - Younger may prefer parent, older may prefer privacy
- - Explain procedures, encourage participation
- - See exam as fun/learning
- - Head-to-toe with genitalia last



Adolescents

- - Exam table, gown, undress in private
- - Confidentiality is critical
- - Offer choice of parent presence
- - Address teen directly
- - Head-to-toe with genitalia last
- - Discuss sexuality, body development, emphasize normalcy



Physical Assessment of the Newborn (General Measurements)

- - Head circumference: 33–35 cm (2–3 cm > chest)
- - Chest circumference: 30.5–33 cm
- - Crown-to-rump: 31–35 cm
- - Head-to-heel: 48–53 cm
- - Birth weight: 2700–4000 g

Newborn Vital Signs

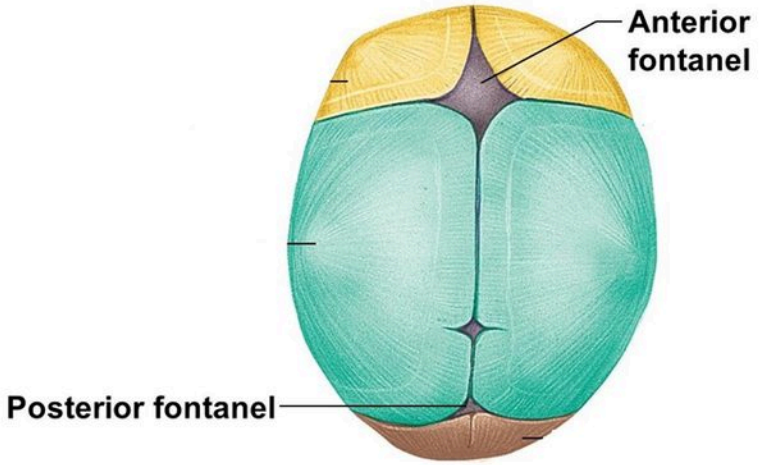
- - Temperature: 36.5–37 °C (Axillary)
- - HR: 110–160 bpm (Apical)
- - RR: 30–60 breaths/min
- - BP: 65/41 mmHg

Newborn General Appearance

- - Flexed posture (head & extremities)
- - Skin: Puffy/smooth at birth, pink/flaky/dry by day 2–3
- - Vernix caseosa, lanugo, acrocyanosis common

Head & Fontanels

- - Anterior fontanel: Diamond-shaped (2.5–4 cm)
- - Posterior fontanel: Triangular (0.5–1 cm)
- - Should be flat & firm

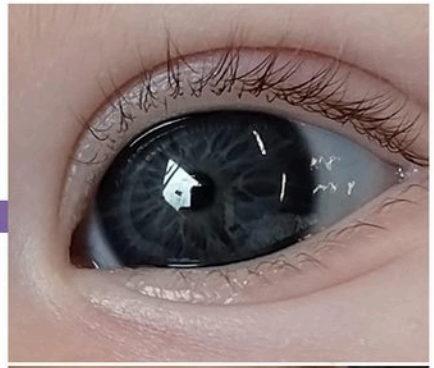


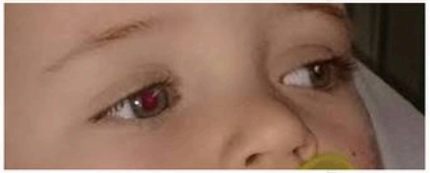
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Figure 5.15a

Newborn Eyes





- a. Lids usually edematous
- b. Iris color—slate gray, dark blue, brown; absence of tears
- c. Presence of red reflex
- d. Corneal reflex in response to touch
- e. Pupillary reflex in response to light
- f. Blink reflex in response to light or touch
- g. Rudimentary fixation on objects and ability to follow to midline

Newborn Ears, Nose

- . Position—Top of pinna on horizontal line with outer canthus of eye
- b. Startle reflex elicited by a loud, sudden noise
- c. Pinna flexible, cartilage present

Nose

- a. Nasal patency
- b. Nasal discharge—Thin, white mucus
- c. Sneezing
- d. Milia

Nose, Mouth & Throat

- - Mouth: a. Intact, high-arched palate
- b. Uvula in midline
- c. Frenulum of tongue
- d. Frenulum of upper lip
- e. Sucking reflex—Strong and coordinated
- f. Rooting reflex
- g. Gag reflex
- i. Absent or minimal salivation
- j. Vigorous cry

Neck, Chest, Back, Rectum



- - **Neck:** Short, thick, tonic neck reflex
- - **Chest:**
- a. Anteroposterior and lateral diameters equal
- b. Slight sternal retractions evident during inspiration
- c. Xiphoid process evident
- d. Breast enlargement
- - **Back:** a. Spine intact; no openings, masses, or prominent curves, b. Trunk incurvation reflex (Galant reflex), c. Anal reflexd. Patent anal opening, Passage of meconium within 48 hours

Extremities



- a. Ten fingers and ten toes
- b. Full range of motion
- c. Nail beds pink, with transient cyanosis immediately after birth
- d. Creases on anterior two thirds of sole
- e. Sole usually flat
- f. Symmetry of extremities
- g. Equal muscle tone bilaterally, especially resistance to opposing flexion
- h. Equal bilateral brachial and femoral pulses

Neuromuscular System

a. Extremities usually maintain some degree of flexion.

- b. Extension of an extremity followed by previous position of flexion
- c. Head lag while sitting, but momentary ability to hold head erect
- d. Able to turn head from side to side when prone
- e. Able to hold head in horizontal line with back when held prone



Thermometer (for measuring temperature)



Stethoscope (for measuring blood pressure and listening to heart, lung, and bowel sounds)



Sphygmomanometer (for measuring blood pressure)



Pocket ruler (to measure size or distance)



Tape measure (to measure circumference and length)



Scale (to measure weight and height)



Gloves (to wear if there is any possible exposure to blood or body fluids)

Equipment of physical assessment

Thank you